

Indian Institute of Technology, Kharagpur



Department of Electronics & Electrical Communication Engineering

Experiment 4: Delta Modulation

Objectives:

1. Design and implement a Delta Modulation Circuit which produces modulated wave as $d(t)$ from the message signal $m(t)$. Also, design and implement a demodulator circuit which will produce $m(t)$ back from $d(t)$.
2. Analyze the effects of step size Δ and oversampling rate on the performance of delta modulation.

Key Components: IC 7474 (D flip flop), IC 710 (differential comparator), IC 356 (JFET operational Amplifier), resistors, capacitors, digital oscilloscope, function generator, breadboard, connecting wires, power supply.

Brief Theory:

Delta Modulation (DM) is a type of analog-to-digital (A/D) and digital-to-analog (D/A) signal conversion technique. It is a simplified form of Differential Pulse Code Modulation (DPCM), where instead of transmitting absolute values of sampled signals, only the difference between consecutive samples is encoded and transmitted. DM significantly reduces bandwidth requirements compared to Pulse Code Modulation (PCM) because it encodes only changes in amplitude rather than absolute amplitude levels. It is widely used in low-bandwidth voice communication and speech processing applications. In Delta Modulation, a one-bit quantizer is used to approximate the input signal using a staircase waveform. The modulator compares the present sample with the previous one and transmits either:

1. A '1' (UP step) if the new sample is higher than the previous sample.
2. A '0' (DOWN step) if the new sample is lower than the previous sample.

The following are some of the features of delta modulation.

1. Delta modulation employs oversampling to produce a high signal-to-noise ratio.
2. An oversampled input is taken to make full use of the signal correlation.
3. The input sequence is much higher than the Nyquist rate.
4. The step size is very small, that is, Δ

Block Diagram of Delta Modulation

The integrator circuit has a T_s delay. The integrator's output is a T_s -delayed staircase approximation. The error signal is the difference between this staircase approximation and the current sampling input signal. This erroneous signal is sent into the quantizer circuit, which is made up of a hard limiter and an input-output relationship. The mistake is quantized here into two values, namely. The quantizer output is then coded to generate the appropriate delta modulated wave. Demodulation is performed in the receiver circuit using an integrator and a low-pass filter. The modulated wave is first decoded with a decoder, and then the staircase approximation is recreated by feeding the positive and negative pulses produced by the decoder to the integrator. Out-of-band quantizing noise from the high-frequency staircase waveform is eliminated by running the signal through a low-pass filter with a bandwidth equal to the original signal bandwidth.

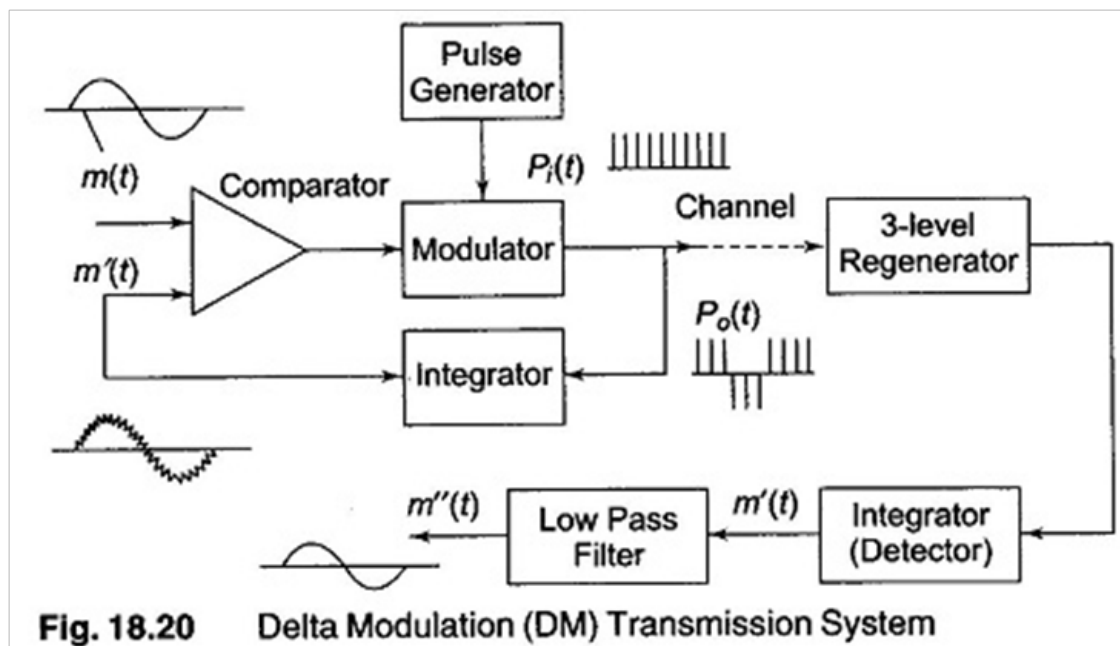
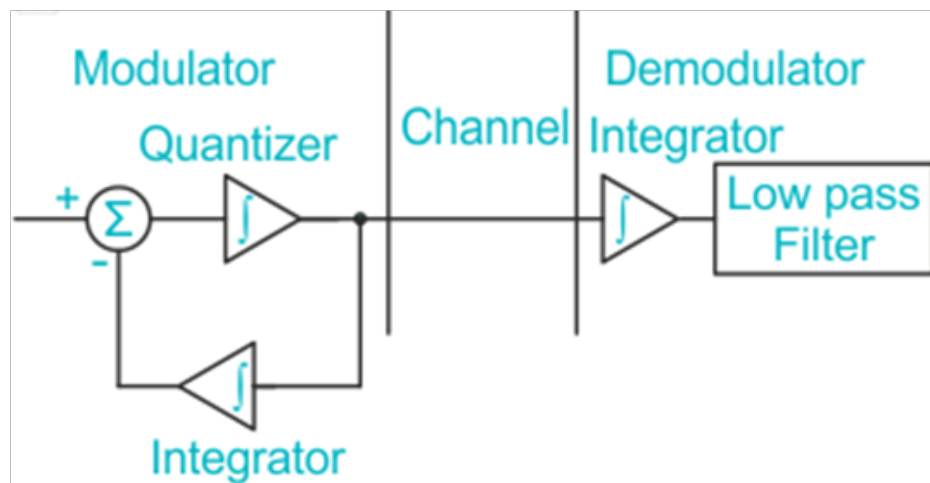


Fig. 18.20 Delta Modulation (DM) Transmission System

Expected Outcomes

1. Observe the effect of step size Δ on the quality of the modulated signal and its susceptibility to slope overload and granular noise.
2. Verify that delta modulation achieves efficient data transmission by reducing bit rate while maintaining signal integrity.